

Data Collection and Ethics

ARBO-WATCH - Capacity Building

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Outline

- 1 Introduction
- 2 Learning Outcomes
- 3 Understanding Data and its Importance
- 4 The data lifecycle
- 5 Principles of good data collection
- 6 Methods of data collection
- 7 Tools for Data Collection
- 8 Data Management Concepts
- 9 Data Quality Management
- 10 Data Security and Ethics

Introduction

- Data collection is a fundamental step in research and decision-making, involving the systematic gathering of information to answer questions, test hypotheses or monitor outcomes.
- Ethics in data collection ensures that the rights, dignity and privacy of individuals are protected throughout the research or operational process.
- This topic explores both the technical aspects of collecting accurate, reliable data and the moral responsibilities involved in handling that data with integrity and respect.

By the end of this training, the participant should be able to:

- 1 Identify the various types of data
- 2 Describe the fundamental principles of data collection
- 3 Compare and apply various data collection methods.
- 4 Assess the quality and integrity of collected data, identifying potential sources of error, bias, or inconsistency.
- 5 Apply ethical principles in data collection and management.

Understanding Data and its Importance

Data refers to raw facts and figures collected from various sources. It can be in the form of numbers, text, images, sounds.

- The number of Chikungunya cases reported in a certain hospital in a span of two months.
- Survey responses from patients in a hospital (Yes/No, Age, Gender, Symptoms).
- Disease prevalence aggregated by administrative units (counties, districts, countries, etc)

- 1 **Quantitative Data:** This is numerical information that is measured or counted and recorded in a variety of forms, including counts, scores or ranks.

Example: Number of confirmed cases (e.g., 50 cases of Chikungunya in Mombasa, Kenya).

- 2 **Qualitative data:** Qualitative data encompasses non-numerical information that describes qualities, characteristics or concepts. It is often collected through methods like interviews, observations or textual analysis.

Example: Beliefs about disease causes (e.g., malaria attributed to witchcraft vs. mosquitoes)

Types of Data II

- ③ **Continuous Data:** Continuous data represents values that can take any value within a given range and can be infinitely divided. It is inherently quantitative and is often represented on a continuous scale, without any gaps or jumps.
Example: Rainfall, humidity and temperature influencing vector-borne diseases (e.g., malaria).
- ④ **Discrete data:** Discrete data is numerical data that has a finite or countable number of possible values.
Example: The number of new borns in a day, at a certain hospital, recorded over time.
- ⑤ **Nominal data:** This data type consists of categories or labels that do not have a natural order or ranking.
Example: Gender (Male (0) , Female (1), Infection status (Positive (1), Negative (0))

Types of Data III

- 6 **Ordinal Data:** Ordinal data is a type of categorical data where the categories have a natural order or ranking. The categories can be arranged in a meaningful sequence such as from low to high or from least to most.

Example: Disease Severity Levels like ; Asymptomatic → Mild → Moderate → Severe (e.g., for COVID-19 or malaria)

- 7 **Interval Scale data:** This is numerical data where:
- The intervals between values are equal and meaningful.
 - But there is no true zero point (zero does not mean absence).
 - Because of this, you can do addition and subtraction, but ratios don't make sense.

Example: Body Temperature ($^{\circ}\text{C}$ or $^{\circ}\text{F}$) - 37°C , 38.5°C , 40°C for patients with malaria or COVID-19. Difference between 37°C and 38°C is meaningful (1°C), but 40°C is not “twice as hot” as 20°C . Also, 0°C does not mean that there is no temperature.

- 8 **Ratio Scale Data:** Ratio scale data has equal intervals and a true zero, where zero implies absence. This is numerical data where:
- Ratios are meaningful (e.g., "twice as many").
 - There is true zero point (zero means absence).
 - One can do addition and subtraction, and the ratios make sense.

Examples:

- Number of new cases reported daily (0 cases = none).
- Viral load (measured in copies/mL; 0 = absence of virus).
- Incubation period in days (0 days = no incubation).
- Duration of hospital stay (days).

- 9 **Time series data / Longitudinal Data** Time series data consists of observations recorded sequentially over regular intervals of time. This type of data is used to track changes or trends in a variable over a specified period.

Example: Number of Chikungunya infections, at a certain hospital, recorded every day for one year.

- 10 **Cross-sectional data** Cross-sectional data captures information from subjects at a single point in time.

Example: Number of Chikungunya infections, at a particular day, for all the hospitals in Kenya at a particular day.

Data Importance I

Data is vital because it serves as the foundation for decision-making, problem-solving and innovation across every field. Here are some key reasons why data is important:

- 1 **Informed Decision-Making** : Reliable data helps organizations and individuals choose the best course of action based on evidence rather than guesswork.

Example: A hospital uses patient data to decide on treatment protocols.

- 2 **Performance Measurement** : Data allows tracking progress and evaluating success over time.

Example: A hospital reviews surgery success rates to enhance procedures and training.

- ③ **Problem Identification and Solutions** : Data reveals trends, patterns and anomalies that can highlight problems and suggest solutions.
Example: Immunization coverage data identifies regions with low vaccination rates, prompting targeted campaigns.
- ④ **Innovation and Improvement** : Data fuels research, product development and service enhancement.
Example: Researchers use clinical trial data and patient registries to develop new drugs, treatments and medical devices.
- ⑤ **Forecasting and Planning** : Historical data is used to predict future trends. Data can show which areas or populations need more resources, such as hospitals, staff or equipment.

The data lifecycle I

- The Data Life Cycle refers to the series of stages that data goes through from initial creation or acquisition to its eventual disposal.
- Understanding this cycle is essential for managing data effectively, ensuring its quality, security, and usability throughout its lifespan.

The data lifecycle II

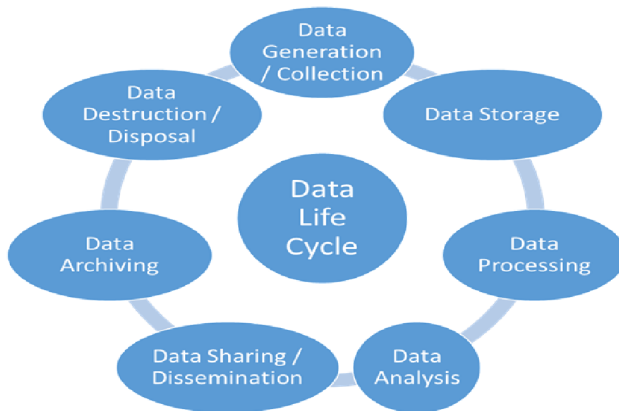


Figure: Data Life Cycle

The data lifecycle III

1. **Data Generation / Collection:** Data is created or collected from various sources such as surveys, sensors, transactions, social media, medical devices, etc. Example: Patient data collected during a clinic visit.
2. **Data Storage:** Data is stored in databases, cloud systems or physical storage. Example: Electronic Health Records (EHR) stored in a hospital database.
3. **Data Processing:** Raw data is cleaned, transformed, and organized for analysis. Example: Removing duplicate patient entries and correcting date formats.
4. **Data Analysis:** Data is analyzed to extract insights, patterns, and trends. Example: Analyzing patient data to detect disease outbreaks.
5. **Data Sharing / Dissemination:** Insights or datasets are shared with stakeholders, researchers, or the public. Example: Ministry of Health sharing health trends with the public.

- 6. **Data Archiving** : Infrequently used data is moved to long-term storage for historical reference. Example: Archiving records of discharged patients for future reference or audits.
- 7. **Data Destruction / Disposal** : Data that is no longer needed is securely deleted or destroyed. Example : Securely deleting old patient files that exceed the legal retention period.

Principles of good data collection I

Effective data collection is essential for research, policy-making, and improving intended outcomes. Here are key principles of data collection, along with practical examples from the health field:

1. **Relevance** : Only collect data that is directly related to the research.

Example: Malaria Surveillance

Relevant data: bed net usage, recent travel to endemic areas, rapid diagnostic test (RDT) results.

Irrelevant data: patient's driving license number.

2. **Accuracy** : Ensure the data collected is correct and reflects the true situation.

Example: Malaria Case Data

Accurate: Using Rapid Diagnostic Test (RDT) results as case confirmation.

Inaccurate: Recording all fever cases as malaria without laboratory confirmation.

Principles of good data collection II

- 3. **Completeness** : All required data fields must be filled without omission.

Example: Ebola Contact Tracing

Complete: Every contact's name, age, sex, relationship to case, address, and daily follow-up status recorded.

Incomplete: Missing contact addresses, making follow-up impossible.

- 4. **Consistency** : Data should be collected and recorded in a uniform manner across time and sources.

Example: Cholera Outbreak Line List

Consistent: All patients' dates are recorded in the same format (DD/MM/YYYY).

Inconsistent: Some records use DD/MM/YYYY, others MM/DD/YYYY, causing confusion.

Principles of good data collection III

- 5. **Timeliness** : Data must be collected and reported within a time frame that makes it useful.

Example: Cholera Outbreak Response

Timely: Health facilities immediately notify local health authorities of suspected cholera cases, allowing rapid water and sanitation interventions.

Untimely: Reports submitted after the outbreak has already spread to neighboring villages.

- 6. **Confidentiality** : Protect the privacy and identity of individuals whose data is being collected.

Example: Ebola Survivor Records

Confidential: Storing survivor health details in encrypted electronic systems.

Breach: Sharing survivor lists with the public or media.

Principles of good data collection IV

- 7. **Validity** : Data should accurately measure what it is intended to measure.

Example : COVID-19 Testing

Valid: Using PCR or approved antigen tests to confirm infection.

Invalid: Counting any patient with a cough as a confirmed COVID-19 case without lab testing.

- 8. **Objectivity** : Data collection must be free from bias or personal opinions.

Example: COVID-19 Case Reporting

Objective: Reporting only laboratory-confirmed positive tests as cases.

Non-objective: Including people “who look sick with cough” without diagnostic confirmation.

Principles of good data collection V

- 9. **Ethical Considerations** : Data must be collected with informed consent and used responsibly.

Example: Malaria Clinical Trials

Ethical: Providing full disclosure of risks and benefits to participants before enrolling in drug trials.

Unethical: Enrolling participants in a trial without explaining potential side effects.

- 10. **Cost-Effectiveness** : Data should be collected efficiently, balancing cost with the value of the information obtained.

Example: Ebola Monitoring

Cost-effective: Training local community health volunteers to track cases and contacts instead of relying solely on costly international staff.

Not cost-effective: Using only expatriate experts for surveillance when local teams can do most of the work with training.

Methods of data collection I

Data collection methods refer to the systematic approaches and techniques used to gather information from various sources for analysis, decision-making and knowledge building. Choosing the right method is crucial, as it determines the accuracy, reliability and validity of the data collected.

1. **Surveys and Questionnaires:** These are standardized tools (paper, electronic, or oral) used to collect data from individuals. E.g: A community health questionnaire to measure awareness of malaria prevention methods.
2. **Interviews:** A qualitative method to gather detailed information through direct interaction with participants. Eg: Conducting one-on-one interviews with cancer survivors to explore their coping mechanisms.

Methods of data collection II

- 3. **Observational Studies:** Involves systematically watching and recording behaviors or events. Eg: Observing how medical staff interact with patients in an emergency department to evaluate the quality of care.
- 4. **Focus Groups:** Group discussions that explore specific topics in depth to gather diverse perspectives. Eg: A focus group with new mothers discussing their postpartum experiences and healthcare needs.
- 5. **Wearable Technology:** Data collected from devices worn by participants, collecting data over time Eg: Using fitness trackers to monitor physical activity levels in a population participating in a lifestyle intervention study.
- 6. **Secondary Data:** Utilizes existing records/data for analysis. Eg: Analyzing hospital discharge records to study the readmission rates of heart failure patients.

Tools for Data Collection

Data collection tools are essential for gathering, analyzing and managing information in research. Below is a list of common tools used for data collection.

1. **Online Survey Platforms:-** Kobocollect, Google Forms , SurveyMonkey , etc
2. **Interview Software:-** Zoom, Skype, Microsoft Teams, etc
3. **Wearable Devices :** - Smart Watch (Apple, Samsung, Oppo, etc), Garmin, Fitbit, etc.
4. **Focus Group Software :-** Miro, FocusVision, etc.
5. **Audio and Video Recording Devices :** - Smartphones, Digital voice recorders, etc.

Data Management Concepts I

Data management refers to the systematic process of collecting, storing, organizing, securing and using data efficiently and responsibly:

1. **Data Collection:** The process of gathering information from relevant sources.
2. **Data Storage:** Saving collected data in a secure and organized form for future use.
3. **Data Quality:** To ensure data quality, one has to make sure data is accurate, complete, consistent and reliable.
4. **Data Security:** Data security is protecting data from unauthorized access, loss or corruption.
5. **Data Privacy:** Information is collected, stored and shared according to consent and legal requirements.
6. **Data Sharing:** Exchanging data between stakeholders while ensuring privacy and security.

Data Management Concepts II

- 7. **Data Analysis:** Examining data to identify trends, patterns and insights.
- 8. **Data Governance:** Establishing policies, standards and responsibilities for data management.
- 9. **Data Archiving:** Storing old or inactive data securely for long-term retention and possible future use.

Data Quality Management

Data Quality Management (DQM) is the systematic process of ensuring that data is accurate, complete, reliable, consistent and timely throughout its lifecycle so it can be effectively used for decision-making.

1. **Accuracy:** - Data correctly represents the real-world values.
2. **Completeness:-** All necessary data is present.
3. **Consistency:-** Data is uniform across systems and time.
4. **Timeliness:-** Data is available when needed and updated.
5. **Validity :-** Data meets the required format and standards. regularly.

1. Data Security

This is the practice of protecting digital and physical data from unauthorized access, alteration, loss or theft.

The key measures to ensure data security include:

- **Access Control:** Only authorized healthcare staff can view or edit patient records.
- **Encryption:** Data is stored and transmitted in coded form to prevent unauthorized use.
- **Backups:** Regularly saving data copies to secure locations to prevent loss.
- **Audit Trails:** Tracking who accessed or changed records.

2. Data Ethics

Data ethics is the set of moral principles and guidelines governing how data is collected, stored, shared and used, ensuring respect for individual rights and societal values.

The key Principles of data ethics include:

- **Informed Consent:** Patients must agree to how their data will be used.
- **Privacy:** Personal health details should not be disclosed without permission.
- **Non-Maleficence:** Avoiding harm through misuse of data.
- **Transparency:** Being clear about how and why data is used.

Q. and A.

Class Exercise

Kindly open the link below and attempt all the questions.

- <https://forms.gle/t1dVxDHz3pkp8aM88>

THANK YOU